DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination-2023

Course: B. Tech. **Semester : III**

Branch : Electronics and Computer Engineering/ Electronics and Computer Science Engineering

Subject Code & Name: Programming, Data Structure and Algorithm using C (BTECPC303)

Max Marks: 60 **Date:06-01-24** **Duration: 3 Hr.**

Instructions to the Students:

1. *All the questions are compulsory.*
2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
3. *Use of non-programmable scientific calculators is allowed.*
4. *Assume suitable data wherever necessary and mention it clearly.*

		(Level/CO)	Marks
Q. 1	**Solve Any Two of the following.**		**12**
A)	What is Linked List? Explain various operations performed on Linked List.	L2/CO1	**6**
B)	Draw a labeled diagram of linked list having 5 nodes; a)Singly linked list b)Singly Circular linked list c)Doubly Circular linked list	L2/CO1	**6**
C)	Write a C program to delete first node of singly linked list.	L1/CO1	**6**
Q.2	**Solve Any Two of the following.**		**12**
A)	Define Stack. Explain working of stack with diagram.	L3/CO2	**6**
B)	Explain Infix, Prefix and Postfix expression.	L2/CO2	**6**
C)	Write a C program to implementation of stack using linked list	L3/CO2	**6**
Q. 3	**Solve Any Two of the following.**		**12**
A)	Define Queue. Explain working of queue with diagram.	L2/CO3	**6**
B)	What is the limitation of simple queue? Explain circular queue.	L3/CO3	**6**
C)	Write a note on i) PRIORITY QUEUE ii)DEQUE	L2/CO3	**6**
Q.4	**Solve Any Two of the following.**		**12**
A)	Explain the linked list representation of Binary Tree	L2/CO4	**6**
B)	Define i) Tree ii) Degree of Node iii) Degree of Tree iv) Graph v) Adjacent Node vi) Weighted Graph	L2/CO4	**6**
C)	State various ways of graph representation. Explain anyone.	L2/CO4	**6**
Q. 5	**Solve Any Two of the following.**		**12**
A)	What is Algorithm? Define Asymptotic notation with example.	L2/CO5	**6**
B)	Explain *Binary Search* with help of following array. *int array[8] = {8,1,3,7,39,27,16,45}*	L3/CO5	**6**
C)	Write an algorithm for selection sort.	L2/CO5	**6**

END